

TATA TECHNOLOGIES LTD.

TECH PULSE FY-26 | AI/ML Capstone Project

PROJECT SYNOPSIS

Student Details

Institute/University: Pimpri Chinchwad College of Engineering (PCCOE), Pune / SPPU

Programme / Batch: B.Tech Computer Engineering / Expected graduation: 2027

Student Name: Aditi Ashok Bachulkar

PRN: 123B1B085

1. Case Study ID and Project Title / Topic

Case Study ID (CS1 to CS5): CS1

Project Title / Topic: AUTOSAR HLD Document Analysis Assistant using RAG

2. Aim

To design and implement an AI/ML-based assistant that uses Retrieval-Augmented Generation (RAG) techniques to analyze an AUTOSAR-style High-Level Design (HLD) document and help users retrieve relevant technical information with supporting source, page, and section citations.

3. Objectives

- To develop an assistant for retrieving relevant information from an AUTOSAR-style HLD document.
- To implement a RAG pipeline for PDF text extraction, chunking, embedding, and retrieval.
- To use a pre-trained sentence-transformer model to generate document and query embeddings.
- To store and retrieve relevant HLD information using a persistent ChromaDB vector database.
- To provide transparent answers with source, page, and section citations for verification.
- To evaluate the system using predefined questions based on retrieval success, citation coverage, and answer coverage.
- To maintain a controlled and explainable prototype using synthetic educational HLD data.

4. Scope

In Scope – To Be Implemented

- Processing a synthetic AUTOSAR-style HLD PDF.
- Extracting text from the document page by page.
- Creating page-aware document chunks.
- Generating embeddings using the all-MiniLM-L6-v2 model.
- Storing embeddings and document metadata in ChromaDB.
- Retrieving relevant HLD sections for user questions.
- Generating extractive responses based on retrieved evidence.
- Displaying source, page, and section citations.

- Providing a Streamlit-based user interface.
- Evaluating the system using a predefined set of 10 questions.

Out of Scope – Will Not Be Implemented

- Use of proprietary or confidential automotive documents.
- Direct integration with real vehicle ECUs or automotive systems.
- Production deployment of the system.
- Complete AUTOSAR configuration analysis.
- Safety, cybersecurity, timing and calibration analysis.
- Automatic and reliable detection of every unsupported question.
- Production-level automotive safety or compliance validation.

5. Proposed Approach

A. Pre-trained Model

The project uses the pre-trained all-MiniLM-L6-v2 sentence-transformer model for generating embeddings of HLD document chunks and user queries. The default demonstration uses an extractive response mode and does not require a generative LLM. An optional OpenAI-compatible API branch is available through environment configuration.

B. Embedding Model

all-MiniLM-L6-v2 using sentence-transformers 6.1.0. The same embedding model is used for both document chunks and user queries.

C. Vector Store

ChromaDB 1.5.9 is used as the persistent vector database. The collection is named hld_chunks, and the data is stored locally in the vector_db directory using cosine similarity.

D. RAG Pipeline

The HLD PDF is first processed page by page using PyMuPDF. The extracted text is divided into page-aware chunks of 850 characters with 150 characters of overlap. Each chunk is converted into an embedding using all-MiniLM-L6-v2 and stored in ChromaDB along with page and section metadata.

When the user enters a question, the question is also converted into an embedding. ChromaDB retrieves the most relevant chunks using cosine similarity. The retrieved evidence is then used to generate an extractive response along with source, page, and section citations.

E. Service Layer / UI

The system uses Streamlit 1.65.0 to provide a simple interactive interface. Users can upload the HLD document, build or rebuild the knowledge base, enter questions, view answers with evidence and citations, and run the predefined evaluation.

Application file: Code\app.py. Run command: streamlit run Code\app.py

F. Tools / Deterministic Components

- PyMuPDF: PDF text extraction.
- SentenceTransformers: Document and query embeddings.
- ChromaDB: Vector storage and similarity-based retrieval.

- Python: Evaluation and processing logic.
- Page and section metadata: Used for citation generation.
- Predefined questions: Used for system evaluation.
- Streamlit: User interface.

6. Proposed System / Architecture

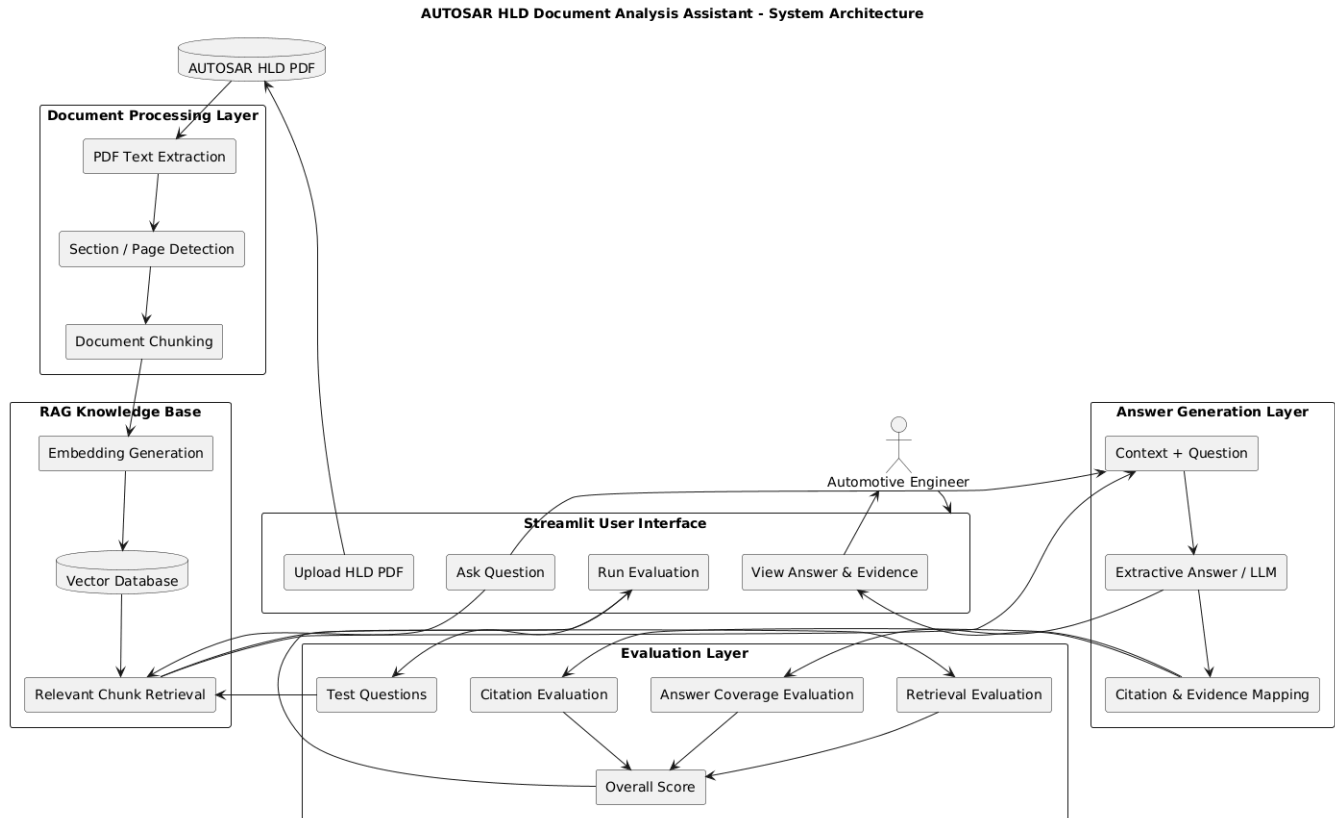


Figure 1: System Architecture of AUTOSAR HLD Document Analysis Assistant

7. Input Data / Knowledge Base

A. Data / Knowledge Sources

The system uses a synthetic educational AUTOSAR-style High-Level Design (HLD) document named `sample_hld.pdf`. The document was created specifically for controlled project demonstration and evaluation and does not contain proprietary or confidential automotive information.

B. Approximate Size and Format

PDF format, 11 pages, 11 page records, 11 chunks, and 11 indexed documents. The verified project configuration does not record the original file size.

C. Preprocessing and EDA Planned

The PDF is processed page by page using PyMuPDF 1.28.2. Text is extracted and cleaned using basic whitespace stripping, and empty pages are ignored. Section information is inferred from the beginning of each page. The document is divided into page-aware character chunks of 850 characters with 150 characters of overlap.

OCR, duplicate removal, and header/footer removal are not implemented in the current prototype.

8. Expected Outcomes and Evaluation Plan

A. Expected Outputs

The expected outputs of the system are:

- A working Streamlit interface for HLD document analysis.
- A persistent ChromaDB knowledge base.
- Retrieval of relevant HLD sections based on user questions.
- Extractive answers based on retrieved evidence.
- Source, page, and section citations.
- Evidence displayed to the user for verification.
- Evaluation results for the predefined 10-question test set.

B. Evaluation Method

The system is evaluated using 10 predefined questions related to the HLD document. Evaluation metrics:

- Retrieval Success Rate: Checks whether at least one retrieved page matches the expected page.
- Citation Coverage: Checks whether the generated answer contains source citation information.
- Answer Coverage: Measures the presence of expected answer keywords in the response.
- Overall Score: Calculated as the average of retrieval success, citation coverage, and answer coverage.

C. Actual Demo Evaluation Results

Demo Extractive Mode Results:

Metric	Result
Retrieval Success Rate	100.0%
Citation Coverage	100.0%
Answer Coverage	71.5%
Overall Score	90.5%
Latency	Not measured

Answer Coverage is based on expected keyword presence and does not represent full semantic accuracy.

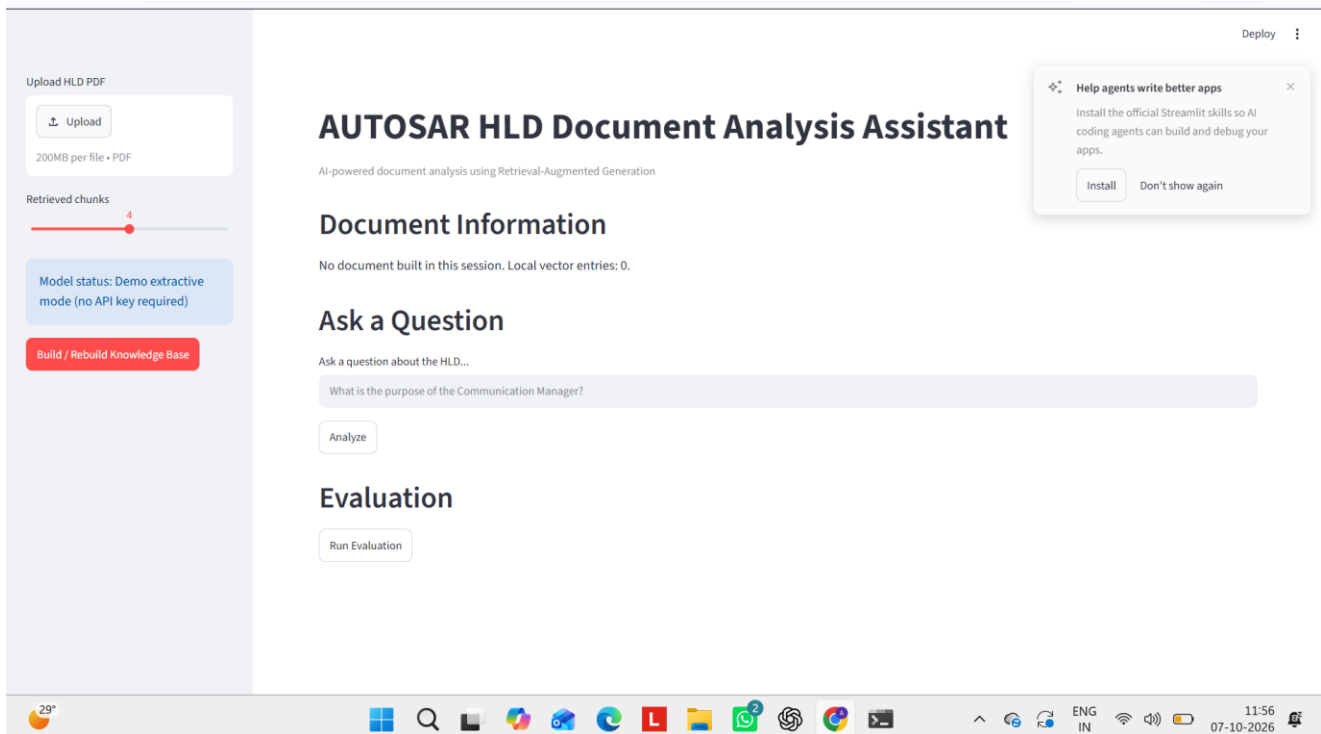


Figure 2: Streamlit User Interface of the Proposed System

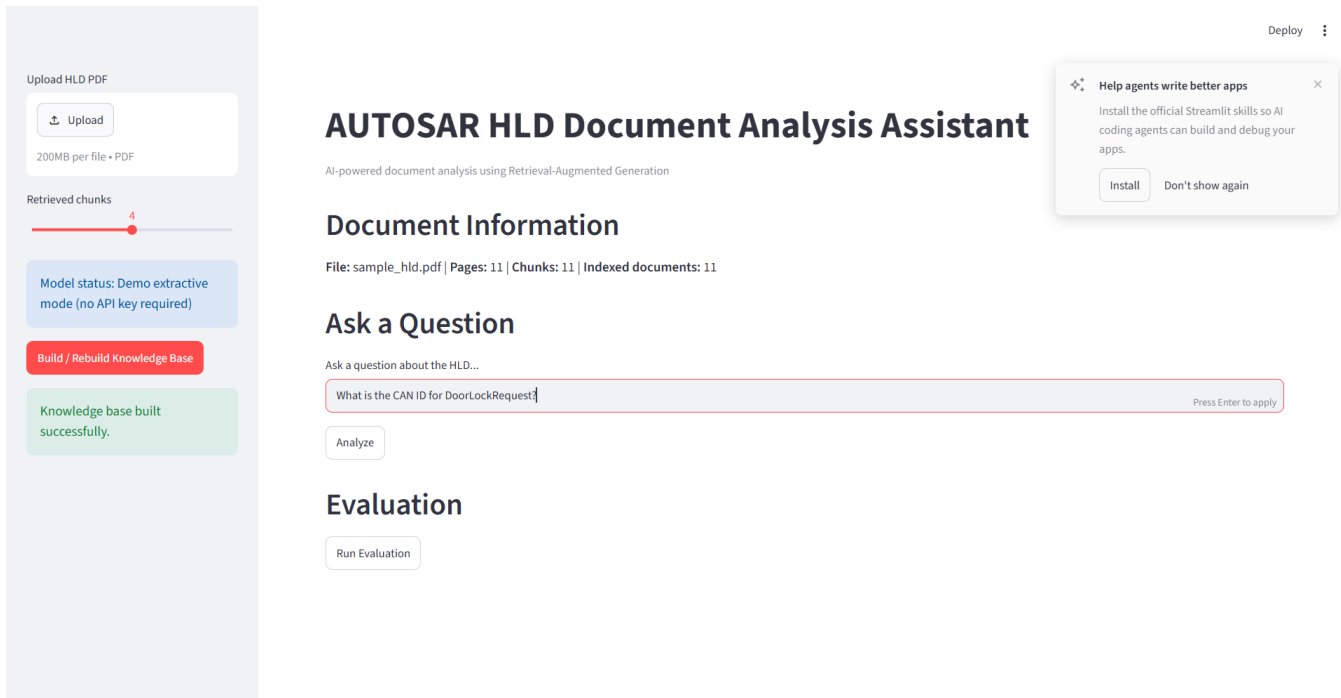


Figure 3: HLD Question, Answer and Source Citations

← → ↺ http://localhost:8501

Deploy

Upload HLD PDF

Upload

200MB per file • PDF

Retrieved chunks

4

Model status: Demo extractive mode (no API key required)

Build / Rebuild Knowledge Base

AUTOSAR HLD Document Analysis Assistant

AI-powered document analysis using Retrieval-Augmented Generation

Document Information

File: sample_hld.pdf | Pages: 11 | Chunks: 11 | Indexed documents: 11

Ask a Question

Ask a question about the HLD...

What is the CAN ID for DoorLockRequest?

Analyze

AI Answer

Based on the retrieved HLD evidence: 5. Communication Architecture Communication Manager coordinates communication between application components and the CAN Interface. It maps application messages to fictional CAN frames and distributes received frames to subscribed components. CAN Interface owns controller access and validates frame timing. Example signals: DoorLockRequest (CAN ID 0x221), CabinSetpoint (CAN ID 0x315), DoorStateStatus (CAN ID 0x222). Communication Manager depends on CAN Interface; application components depend on Communication Manager rather than directly accessing CAN. Synthetic demonstration document | Aurora Platform

[Source: sample_hld.pdf, Page 6, Section: 5. Communication Architecture] [Source: sample_hld.pdf, Page 4, Section: 3. Door Control Component] [Source: sample_hld.pdf, Page 7, Section: 6. Sensor and Actuator Services] [Source: sample_hld.pdf, Page 9, Section: 8. Functional Flows]

Response mode: demo

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Figure 4: HLD Question and Extractive Answer with Source Citations

← → ↺ http://localhost:8501

Deploy

Upload HLD PDF

Upload

200MB per file • PDF

Retrieved chunks

4

Model status: Demo extractive mode (no API key required)

Build / Rebuild Knowledge Base

File: sample_hld.pdf | Pages: 11 | Chunks: 11 | Indexed documents: 11

Ask a Question

Ask a question about the HLD...

What is the CAN ID for DoorLockRequest?

Analyze

Evaluation

Run Evaluation

```
{
  "questions" : 10
  "retrieval_success_rate" : 100
  "citation_coverage" : 100
  "answer_coverage" : 71.5
  "overall_score" : 90.5
}
```

Prototype metrics: retrieval/page matching, citation presence, and keyword coverage. They are not a substitute for human evaluation.

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Figure 5: Evaluation Results of the Proposed System

9. Responsible AI Measures

- Synthetic educational data is used instead of proprietary or confidential automotive data.
- Retrieved evidence is displayed to make the response traceable.
- Source, page, and section citations are provided with responses.
- The default demonstration does not require an external API key.
- Optional API credentials are stored through environment variables instead of being hard-coded.
- Human review is required before using retrieved information for engineering decisions.
- The system does not claim production-level automotive safety, cybersecurity, or compliance validation.
- Unsupported questions may still retrieve the closest available chunks because the current prototype does not use a similarity threshold.

10. MLOps / Deployment Elements

Basic reproducibility is supported through requirements.txt, configuration files, project documentation, and saved evaluation outputs. The current prototype uses local ChromaDB storage and is designed for local demonstration through Streamlit.

Docker-based deployment, experiment tracking, cloud deployment, authentication, and production monitoring are not implemented in the current version.

11. Planned AI-Tool Usage

AI Tool / Model

AI-assisted development and documentation tools

Purpose

Used for project planning, technical guidance, coding assistance, debugging, troubleshooting, and refinement of project content.

Artifact Affected

Python source code, configuration files, evaluation workflow, project documentation.

Student Declaration

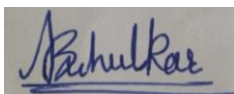
I will not change the approved case-study scope, major methodology, or implementation approach without prior faculty approval.

I confirm that the work submitted under this synopsis represents my own understanding and contribution, and I will be responsible for explaining every significant part of the selected case study and its implementation.

Name: Aditi Ashok Bachulkar

PRN: 123B1B085

Signature:



Date: 07/09/2026